

# Poorvi Hosakere Chandrashekhar

GitHub, LinkedIn, Google Scholar

+91 7019137369

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|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Education               | <b>Carnegie Mellon University</b><br>Master of Computational Data Science (MCDS)<br><br><b>International Institute of Information Technology - Hyderabad</b><br><i>Dual Degree Program:</i> B.Tech in CS and MS by Research in CNS<br><b>Master of Science</b> , Computational Natural Sciences<br><b>Bachelor of Technology with Honors</b> , Computer Science and Engineering<br><b>Admits:</b> CMU MCDS, ETH MSc CSE, CMU MSCB, John Hopkins Biomedical Engineering, Georgia Tech MSCS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Aug 2026 – Dec 2027</b><br><br>Nov 2021 – May 2026<br>Overall GPA: 8.8/10.0                                            |
| Scholastic Achievements | <b>Google Women Engineers (WE) Scholar:</b> Selected as a <b>Top 100 Scholar nationwide</b> for a 2-year technical mentorship program by Google; awarded a scholarship of INR 100,000.<br><b>Uber She++ (UberSTAR):</b> Selected as one of the <b>top 35 women engineers</b> from Tier 1 engineering schools in India for the coveted UberSTAR internship program.<br><b>KVPY Fellow (All India Rank 298):</b> Awarded the prestigious Kishor Vaigyanik Protsahan Yojana fellowship by the Dept. of Science and Technology, Govt. of India (Administered by IISc Bangalore) for aspiring researchers. <b><i>Awarded to the top 1000 students. (0.6% selection rate)</i></b><br><b>Megathon Winner:</b> Secured 1st place (Team of 5) in Hyderabad's largest student-run hackathon ( <b>Prize: INR 30,000</b> ).<br><b>Indian Olympiad Qualifier in Biology (IOQB):</b> Achieved state-level distinction as one of the <b>Top 13</b> students in Karnataka selected for the prelims.                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                           |
| Publications            | <b>HC Poorvi</b> , PK Vinod. (2025) Integrated Cross-Organ Transcriptomic Analysis Uncovers Conserved Gene Signatures Predictive of Allograft Rejection. <i>Accepted at PLOS ONE</i> . <b>bioRxiv Paper Link</b><br><br>Gaurav Bhole*, <b>HC Poorvi*</b> , JR Madhav*, Prabhakar Bhimalapuram, PK Vinod. (2025) Breast Cancer Subtyping with HyperCLSA: A Hypergraph Contrastive Learning Pipeline for Multi-Omics Data Integration. <i>Accepted at 11th International Conference on Pattern Recognition and Machine Intelligence</i> . <b>[Oral Presentation]</b> <b>bioRxiv Paper Link</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                           |
| Research Experience     | <b>Research Assistant</b><br>May 2024 – Present<br><br><b>Lead Researcher &amp; First Author:</b> Integrated Cross-Organ Transcriptomic Analysis (Manuscript accepted at <i>PLOS ONE</i> journal). <ul style="list-style-type: none"><li>Conceptualized a unified molecular framework for allograft rejection by integrating transcriptomic data from <b>672 biopsies across liver, kidney, and heart</b>, challenging traditional organ-specific diagnostic paradigms.</li><li>Constructed organ-specific gene co-expression networks using <b>Leiden clustering</b> and <b>Jaccard similarity</b> to map conserved regulatory modules, identifying a consensus <b>six-module immune cascade</b> (172 genes) capturing the alloimmune response hierarchy.</li><li>Uncovered a novel, highly conserved <b>24-gene cell cycle signature</b> upregulated in rejecting allografts, establishing cellular proliferation as a core, previously underappreciated mechanism of rejection pathology.</li><li>Engineered a machine learning pipeline with <b>SHAP consensus feature ranking</b> to distill high-dimensional data into a robust <b>20-gene universal biomarker panel</b> (<math>AUC &gt; 0.96</math>)</li><li>Validated signatures on an independent lung transplant cohort (<math>n = 243</math>, <math>AUC &gt; 0.97</math>), benchmarking against established RATs panels.</li></ul> | Centre for Computational Natural Sciences and Bioinformatics ( <b>CCNSB</b> ) Lab, <b>IIIT-H</b><br>Advisor: Dr. Vinod PK |

**Research Assistant**  
Jan 2025 – July 2025

**CCNSB Lab, IIIT-H**  
Advisors: Dr. Vinod PK & Dr. Prabhakar Bhimalapuram

**Co-First Author:** HyperCLSA (Accepted at *PREMI 2025* for **Oral Presentation**; to appear in **Springer LNCS Series**).

- Co-developed **HyperCLSA**, a novel deep learning framework for multi-omics breast cancer subtyping, integrating **hypergraph convolutional networks (HGNN)** with **supervised contrastive learning** to align heterogeneous latent spaces.
- Engineered a robust pipeline using **Boruta feature selection** and **radius-based hypergraph construction** to capture high-order gene correlations, followed by **multi-head self-attention** for adaptive modality fusion.
- Achieved state-of-the-art **90.1% accuracy** on the **TCGA-BRCA dataset** ( $n = 850$ ), significantly outperforming established graph-based baselines (MOGONET, MORE) in PAM50 subtype classification.

**Research Assistant**  
July 2025 – Present

**Cognitive Science Lab, IIIT-H**  
Advisor: Dr. Bapi Raju

**Project:** Investigating Structural-Functional Coupling in Task-Based fMRI via Graph Diffusion Wavelets.

- Investigating the generalizability of the **Graph Diffusion Wavelet** framework (originally developed for resting-state SCFC mapping) to task-specific fMRI contexts.
- Testing the hypothesis that **ROI-specific diffusion scales** – which govern information flow on the structural connectome—adapt dynamically under varying cognitive loads (e.g., language processing, motor tasks).
- Benchmarking model performance on the **Human Connectome Project (HCP)** dataset to analyze deviations in linear structure-function coupling across distinct brain states.

Industry  
Experience

**Software Engineering Intern**, Bengaluru, India  
May 2025 – July 2025

**Stripe, Inc**  
Ruby, Typescript, MongoDB, Jest

Developed an end-to-end, user-facing multi-currency charge lookup flow within the Support Chatbot to automate transaction inquiries.

- Productionized the feature to 100% of global users, reducing support case creation by **~120/week** by enabling self-resolution of unrecognized transactions.
- Utilized state-driven workflows to securely handle **sensitive financial data**, ensuring compliance with enterprise security standards.
- Ensured system reliability by implementing comprehensive unit tests (~90% coverage) and authoring technical documentation to bridge knowledge gaps.
- **Received Return Offer** for a full-time Software Engineer role based on exceptional performance and successful feature deployment.

**Software Engineering Intern**, Hyderabad, India  
May 2024 – July 2024

**Uber Technologies, Inc.**  
GO, Typescript, Redux, Fusion.js, Jest

Engineered a full-stack, agent-facing central dashboard using Fusion.js (React) and Go to streamline global entity verification processes.

- Architected a **configuration-driven UI framework** where JSON definitions dynamically render views, eliminating the need for hardcoded region-specific logic.
- **Drastically reduced engineering overhead** for new market launches by abstracting complex regulatory requirements into simple boolean flags, enabling rapid scaling across user, role and environment types.
- **Received Return Intern Offer** based on successful deployment of the dashboard and technical excellence.

Teaching  
Experience

**Head Teaching Assistant, Machine Learning in Natural Sciences (SC4.411):** Spring 2026.  
Instructors: Prof. Vinod PK, Prof. Prabhakar Bhimalapuram.

**Head Teaching Assistant, Computer Programming (CSO.101):** Monsoon 2025.  
Instructors: Prof. Abhishek Deshpande, Prof. Vineet Gandhi, Prof. Girish Varma.

**Teaching Assistant, Statistical Methods in AI (CS7.403):** Spring 2025.

Instructor: Prof. Vineet Gandhi.

**Teaching Assistant, Computer Programming (CSO.101):** Monsoon 2024.

Instructors: Prof. Girish Varma, Prof. Abhishek Deshpande.

**Teaching Assistant, Intro to Software Systems (CS6.201):** Spring 2024.

Instructors: Prof. Anoop Namboodiri, Prof. Prakash Yalla.

**Teaching Assistant, Algorithm Analysis and Design (CS1.301):** Monsoon 2023.

Instructor: Prof. Suryajith Chillara.

## Course Projects

### CS4.409 Data Foundation Systems Project

Aug 2023 – Dec 2023

Indian Driving Dataset (IDD) Filtering Engine

iHub-Data (DST, Govt. of India)

**Engineered a semantic search and filtering platform** using the MERN stack, enabling researchers to query massive Indian Driving Datasets (IDD) based on granular metadata (object classes, image dimensions, scene tags).

- **Architected high-performance retrieval pipelines** by integrating **Elasticsearch** for sub-second metadata querying and **MinIO** for scalable object storage, allowing users to generate and download custom dataset subsets.
- Collaborated with iHub-Data stakeholders to refine requirements, demonstrated the tool to multiple research labs, and successfully deployed the system on the official Data Foundation portal.

### CS3.301 Operating Systems and Networks

Aug 2023 – Nov 2023

#### Project

Distributed Network File System & C-Shell

**Built core system infrastructure from scratch** using C and TCP/IP socket programming to replicate fundamental OS functionalities.

- **Network File System (NFS):** Architected a resilient distributed file system implementing a central Naming Server and concurrent Storage Servers to streamline file operations across a distributed network.
- **Custom C-Shell:** Designed a modular Unix shell supporting advanced command parsing, I/O redirection, piping, and signal handling, replicating core Bash functionality.
- **Optimization:** Executed concurrent client access handling and optimized search mechanisms using semaphores, improving overall system throughput.

## Leadership

Women in Tech Club (LeanIn), *Coordinator*

Aug 2024 – Sep 2025

Entrepreneurship Cell (E-Cell), *Vice Coordinator*

Aug 2023 – Aug 2024

Entrepreneurship Cell (E-Cell), *Content Head*

Aug 2022 – Aug 2024

IIT-H College Fest (Felicity), *Content Team Lead*

Jan 2024 – Apr 2024

Counseling Cell, *Apex Mentor (Freshmen Coordinator)*

Aug 2022 – Aug 2023

Christel House India NGO (Nokia Partnership Project), *Volunteer Tutor*

Jan 2022 - Aug 2022

## Skills

**Languages:** C, Python, R, C++, Ruby, Go, SQL, Bash, Assembly,  $\text{\LaTeX}$ .

**Web Development:** HTML, CSS, JavaScript, TypeScript, React.js, Redux, Node.js, Express.js, Fusion.js, PHP, Bootstrap, MERN Stack.

**Applications & Databases:** Git/GitHub, Vi/Vim, VS Code, MongoDB Atlas, Elasticsearch, MinIO.

**Operating Systems:** Linux (Ubuntu), macOS, Windows, Android.